Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A bundle of parallel rays passes through a lens and becomes 3 times as wide farther beyond it, with rays spreading immediately after the lens. Identify the lens type and sketch the rays.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: d = vΔt; Δt = d/v.
Use the supplied wave speed for its medium and wave type. These classroom models are not real hazard forecasts.

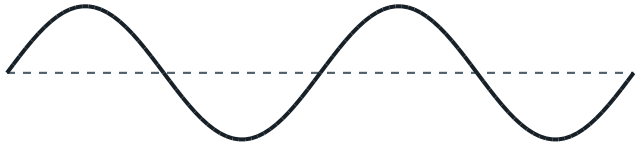
Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

Marzano check - circle: ① ② ③

03 Look and explain

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Build the solution

A toy wave travels at 20 m/s across 200 m. Find travel time.

Marzano check - circle: ① ② ③

Mechanical waves and distant effects

antocciphysics

Continue down each column.

05 Use the idea

Why does knowing wave arrival time alone not establish local damage? Name other evidence.

Marzano check - circle: 1 2 3

06 Choose a method

A wave enters a medium where its speed is 4 times as large, at a small angle below critical conditions. Does the transmitted ray bend toward or away from the normal?

Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, distance doubles. How does travel time change?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two sites have amplitudes 3 and 1 units. Compare relative intensities and name one real-world factor besides distance that could affect the readings.

Marzano check - circle: 1 2 3